

Emmanuel Hernandez

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EDUCATION

University of California, Los Angeles

Los Angeles, CA

Bachelor of Arts in Linguistics and Computer Science

Expected Graduation: June 2028

- **Relevant Coursework:** Data Structures & Algorithms, Algorithms & Complexity (in progress), Operating Systems, Computer Organization, Database Systems, Programming Languages, Software Construction & Tools, Computational Linguistics (in progress), Discrete Mathematics, Linear Algebra, Probability & Statistics, Multivariable Calculus

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, TypeScript, SQL, Swift, Rust

Frontend: React, React Native, Next.js, Expo, Tailwind CSS, HTML/CSS

Backend: Node.js, Express, REST APIs, Firebase, Firestore, PostgreSQL, Prisma, Supabase, MongoDB, JWT auth

Tools: Git, Linux/Unix, Docker, AWS (S3), CI/CD, Vercel, Render, Postman, Google Apps Script, pytest

Agentic: Claude Code, MCP servers, custom subagents & skills, Ollama, LM Studio, Gemini API, LLM eval harnesses

EXPERIENCE

Software Engineering Intern

June 2026 – Aug 2026

Rise the Fenua

Los Angeles, CA

- Built a dashboard (Next.js, TypeScript, Supabase) used by 20+ volunteers to streamline merchandise fulfillment
- Engineered a data pipeline with 6+ SQL migrations to normalize Google Form responses, cutting manual processing by 40%
- Led operations across software, merchandise logistics, and fundraising that raised \$25,000 in donations
- Automated inventory and donation tracking with Google Apps Script, replacing manual shipment record-keeping

Software Engineer Intern

Dec 2025 – June 2026

We Explore Earth

Los Angeles, CA

- Developed features for a real-time event platform built for 20,000+ users, on a 15-member team with 5 designers
- Took the events page and RSVP flow from 0 to 1 across 11 merged PRs, from first skeleton to admin-only delete
- Built the calendar, event card, event detail and attendee-summary components in React Native from Figma designs
- Wrote Express/Firestore endpoints for My Events and event deletion, cascading RSVP cleanup to every attendee

PROJECTS

Database Crash Simulator | C, TypeScript, Bun, Rust, Linux | [GitHub](#)

Aug 2026

- Independently reproduced a permanent data-loss bug in redb, a production Rust database, across 10,942 crash states
- Built a crash simulator that replays every legal post-crash disk state on real ext4/xfs/btrfs images through DB recovery
- Wrote an LD_PRELOAD C shim giving multithreaded syscalls a lock-free total order via one atomic shared-memory counter
- Cut false positives to zero by triaging all 9 flags: 7 bugs in my own tool, 1 planted control bug, 1 real redb bug

LLM Code-Switching Benchmark | Python, pydantic, pytest, Ollama

Aug 2026 – Present

- Built a 402-item benchmark testing whether LLMs follow Matrix Language Frame theory in Spanish-English code-switching
- Found 2 of 3 local LLMs' repair scores fell from 0.79-1.00 to 0.24-0.39 when an English verb sat in a Spanish sentence
- Built an eval harness with bootstrap confidence intervals and a hash-keyed cache, making rescoring free (177 tests)

SwipeBite | Next.js, React, Express, Prisma, PostgreSQL | [Live](#) | [GitHub](#)

Feb 2026 – Present

- Built and deployed a Tinder-style restaurant discovery app (Vercel + Render) with rotating-refresh-token JWT auth
- Wrote a hybrid recommender (content-based, collaborative filtering, Thompson Sampling) that re-ranks the feed per swipe
- Proxied Google Places through the Express 5 + Prisma/PostgreSQL API to keep keys off the client, covered by 153 tests

Physics Remeshing Engine | C++

Jan 2025 – Apr 2025

- Built a C++ remeshing engine solving a second-order ODE by Euler's method over a spring-mass damping model
- Constrained interior points to rectangular, triangular and L-shaped regions using virtual boundary charges
- Selected to present at 2 undergraduate research conferences, including one hosted at UC Berkeley

LEADERSHIP & ORGANIZATIONS

Full-Stack Developer

Sep 2025 – Present

LA Blueprint

Los Angeles, CA

- Selected at a 4% acceptance rate for a student org building full-stack products for LA nonprofits
- Built and maintained React, Node.js/Express, and Firebase features shipped through reviewed pull requests
- Presented architecture proposals and ran sprint planning and design reviews across project teams
- Scoped web applications with prospective nonprofit partners, translating their needs into project requirements

ADDITIONAL

Awards: Dean's List; Southern California Research Symposium; Bay Area Honors Undergraduate Research Conference

Interests: Double Bass (music theory & performance), School on Wheels tutor